IEOR 221 - HW 9 - Jie Wen

1

Let A denote the (arithmetic) average over the sampling dates. For any random variable X, the identity

$$(X-K)^+ - (K-X)^+ = X - K$$

implies a put–call parity at time t for European options written on X:

$$\text{Call}(X,K) - \text{Put}(X,K) = e^{-r(T-t)}\big(\mathbb{E}[X] - K\big). \quad \text{(risk-neutral)}$$

This observation is standard and applies equally to Asian options (with $X = A$).

Apply it three times:

1. **Average-price pair** (underlying A):

$$c_1 - p_1 = e^{-rT}\big(\mathbb{E}[A] - K\big). \quad \text{(average-price Asian definitions)}$$

2. **Average-strike pair** (underlying S_T with strike A):

$$c_2 - p_2 = e^{-rT}\big(\mathbb{E}[S_T] - \mathbb{E}[A]\big). \quad \text{(average-strike Asian definitions)}$$

3. **Vanilla pair** (underlying S_T with fixed strike K):

$$c_3 - p_3 = e^{-rT}\big(\mathbb{E}[S_T] - K\big). \quad \text{(classic put–call parity)}$$

Now add (1) and (2) and subtract (3):

$$(c_1 - p_1) + (c_2 - p_2) - (c_3 - p_3) = e^{-rT}\Big[(\mathbb{E}[A] - K) + (\mathbb{E}[S_T] - \mathbb{E}[A]) - (\mathbb{E}[S_T] - K)\Big] = 0.$$

Rearrange:

$$c_1 + c_2 - c_3 = p_1 + p_2 - p_3.$$

That's exactly the claimed relation.

2

Under Black-Scholes with continuous dividend yield q, its value is

$$\text{Price} = He^{-rT}N(d_2), \quad d_2 = \frac{\ln(S_0/K) + \left(r - q - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}},$$

where $N(\cdot)$ is the standard normal CDF.

Plugging in $S_0 = 960, K = 1000, r = 0.08, q = 0.03, \sigma = 0.20, T = 0.5$:

- $d_2 = \frac{\ln(960/1000) + (0.08 - 0.03 - 0.5 \cdot 0.2^2) \cdot 0.5}{0.2\sqrt{0.5}} \approx -0.1826.$
- $N(d_2) \approx 0.4276.$
- Discount factor $e^{-rT} = e^{-0.04} \approx 0.9608.$

$$\text{Value} = 100 \times 0.9608 \times 0.4276 \approx \$41.08.$$

3

Here's the perpetual-American-put result under GBM with continuous dividend yield (q). The stationary BSM ODE has power-law solutions with exponents $\lambda_1 > 1$ and $\lambda_2 < 0$. For a perpetual **put**, the value in the continuation region ($S \geq S_-$) is

$$V(S) = AS^{\lambda_2}, \quad \lambda_2 = \frac{-(r - q - \frac{1}{2}\sigma^2) - \sqrt{(r - q - \frac{1}{2}\sigma^2)^2 + 2r\sigma^2}}{\sigma^2},$$

with the smooth-pasting exercise boundary

$$S_- = \frac{\lambda_2 K}{\lambda_2 - 1},$$

and closed-form price

$$V(S) = \frac{K}{1 - \lambda_2}\left(\frac{\lambda_2 - 1}{\lambda_2}\right)^{\lambda_2}\left(\frac{S}{K}\right)^{\lambda_2} \quad \text{for } S \geq S_-, \quad V(S) = K - S \text{ for } S \leq S_-.$$

Plugging in the given numbers ($S = 100, K = 80, r = 5\%, q = 1\%, \sigma = 30\%$):

- Compute λ_2:
  $\lambda_2 = -1.0000$. Hence $S_- = \frac{-1}{-2}K = 40$. Since $S = 100 > S_-$, the option is in the continuation region.
- Price:

$$V(100) = \frac{K}{1 - \lambda_2}\left(\frac{\lambda_2 - 1}{\lambda_2}\right)^{\lambda_2}\left(\frac{S}{K}\right)^{\lambda_2} = \frac{80}{2} \cdot \left(\frac{2}{1}\right)^{-1} \cdot \left(\frac{100}{80}\right)^{-1} = 16.00.$$

- Delta (for $S \geq S_-$): $\Delta = \frac{\partial V}{\partial S} = \lambda_2 \frac{V}{S}$. With $\lambda_2 = -1$,

$$\Delta = -\frac{16}{100} = -0.16.$$

- Vega: $\nu = \partial V / \partial \sigma$. There's no tidy closed form because λ_2 depends on σ, but numerically (central difference around $\sigma = 0.30$)

  $$\nu \approx 92.59 \text{ per unit of volatility.}$$

  Many desks quote vega per 1 vol-point (1% absolute); that is

  $$\nu_{1\%} \approx 0.926 \text{ per } 1\sigma.$$